# RESTRICTION TO A LOCALIZED SITE PRESERVES K-INJECTIVITY

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's own argument: it names the general criterion and the exact left adjoint that make it applicable, and refers to the preceding discussion of the chapter for the adjunction itself. All of that is kept verbatim inside the counted proof span.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- Restriction to a localized site and its exact left adjoint. Same chapter, the restriction functor on modules to the localized site and the extension by zero functor, which is left adjoint and exact. Supplies the adjoint pair to which the general criterion is applied.
- Preservation of K-injective complexes by right adjoints of exact functors. Derived Categories, the criterion that a right adjoint of an exact functor preserves K-injective complexes. Supplies the general result invoked in the proof.

*Source reference.* Original lemma 08BJ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4323–4323. Navigation pointer invoked at proof line 4323; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $U$ be an object of $\mathcal{C}$. The restriction of a K-injective complex of $\mathcal{O}$-modules to $\mathcal{C}/U$ is a K-injective complex of $\mathcal{O}_U$-modules.*

*Proof.* Follows immediately from Derived Categories, Lemma 08BJ and the fact that the restriction functor has the exact left adjoint $j_!$. See discussion above. □

*Date*: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.